

Project Phoenix: A Framework for Agentic AI Self-Correction

ABSTRACT :

Team No: 17

Problem Statement:

The current paradigm of Agentic AI is constrained by a fundamental limitation: paradigmatic brittleness. Agent logic is predominantly static, rendering systems vulnerable to catastrophic failure when encountering novel scenarios or edge cases not anticipated during initial design. This lack of autonomous introspection and runtime metaprogramming precludes their deployment in high-stakes, dynamic environments where reliability is non-negotiable. The core scientific challenge is to transition these systems from static, brittle tools into dynamic, anti-fragile agents capable of self-healing, which can adapt and overcome unforeseen errors.

Description:

Project Phoenix introduces a framework for AI to achieve runtime self-correction through a closed-loop process. First, an **Observer** module detects a system failure. This error is passed to a **Critic & Programmer** core, where an LLM generates a corrective code patch. This patch is then rigorously tested in a secure **Validation Gauntlet**. Upon successful validation, an **Integration** module "hot-swaps" the flawed code with the new version. This cycle fosters emergent resilience, making the agent more robust with each successful correction.

Objectives:

- **Build a Baseline Agent & Failure Environment:** Create a sophisticated agent and a benchmark of bugs for evaluation.
- **Engineer a Repair Pipeline:** Develop the Observer, Programmer, Sandbox, and Integration modules.
- **Construct a Secure Validation Gauntlet:** Use containerization to ensure the integrity of the self-correction process.
- **Empirically Validate Efficacy:** Measure performance based on repair success rate, time-to-remediation, and regression avoidance.
- **Introduce Meta-Learning:** Create a knowledge base of successful failure-patch pairs to accelerate future repairs.
- **Establish a Foundation for Future Research:** Publish a research paper and a working prototype to advance self-improving AI systems.

22BQ1A05J5 - Shaik Balaji Mahammad Rafi

22BQ1A05K2 - Shaik Rasheed

22BQ1A05L4 - Thatikonda Devi Sri Chaitanya

Signature of the Guide

(Dr Srinivasa Rao Pendela , M.Tech, Ph.D)